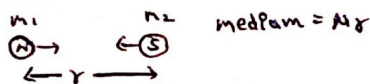


Coulomb's law for Magnetic Forces :

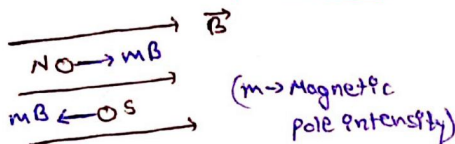
→ Magnetic monopoles never exist in nature.



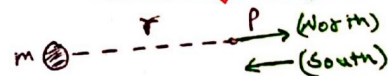
$$F_m = \frac{K m_1 m_2}{r^2} = \frac{\mu_0 \mu_r}{4\pi} \frac{m_1 m_2}{r^2}$$

(m_1, m_2) → Magnetic pole strength (Ampere-m)

Magnetic force on the pole in MF :

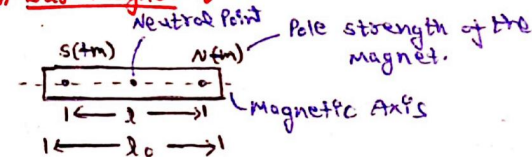


MF due to a Magnetic pole :



$$B = \frac{K m}{r^2} \quad \vec{B} = \frac{K m}{r^2} \cdot \vec{r}$$

Bar Magnet :



→ Poles are not at extreme ends of magnet and situated a bit far from the ends.

→ Hence effective length (l) of bar magnet is less than its geometrical length (l0)

$$l = 0.91 l_0$$

→ Pole strength is directly proportional to its cross section area.

$$m \propto A$$

→ Inside magnet, $\vec{B}$ is straight line.

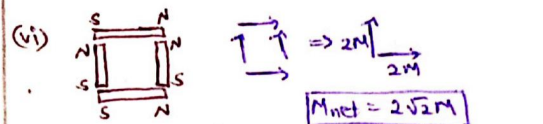
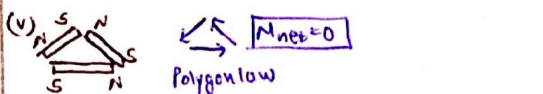
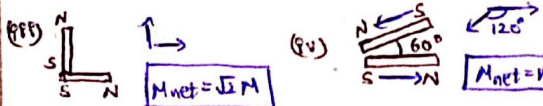
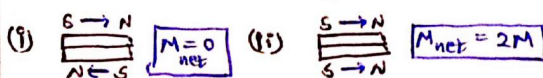


→ Magnetic dipole moment,

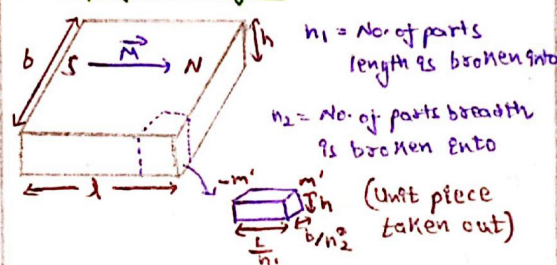
$$\vec{M} = m \cdot \vec{l} \quad (\text{dir}^n \text{ South} \rightarrow \text{North})$$

Magnetism & Matter

Eg: Find net $\vec{M}$ of system of following identical bar magnets.



Cutting of Bar Magnet :



For this small piece

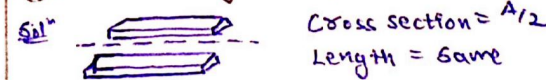
$$A_{\text{new}} = \frac{b \cdot h}{n_2} \quad M' = \frac{m}{n_2}$$

$$M_{\text{new}} = m' \left(\frac{l}{n_1} \right) = \frac{m l}{n_1 n_2}$$

$$\Rightarrow M_{\text{new}} = \frac{M}{n_1 n_2}$$

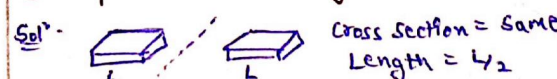
Eg: Bar magnet cut into equal parts

(i) Parallel to length



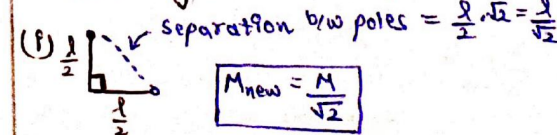
$$M_{\text{new}} = \frac{M}{2}$$

(ii) Perpendicular to length

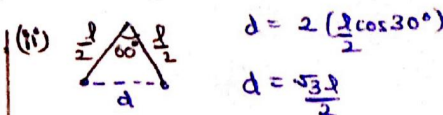


$$M_{\text{new}} = \frac{M}{2}$$

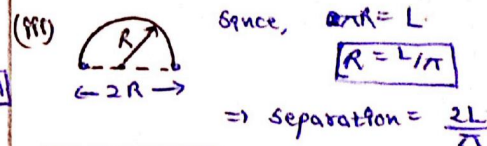
Eg: Bar magnet of length l & magnetic moment M bent as following. Find new magnetic moment :



$$M_{\text{new}} = \frac{M}{\sqrt{2}}$$



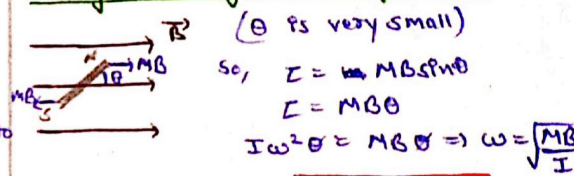
$$M_{\text{new}} = \frac{M \sqrt{3}}{2}$$



$$M_{\text{new}} = \frac{2M}{\pi}$$



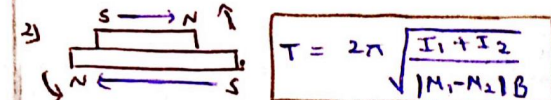
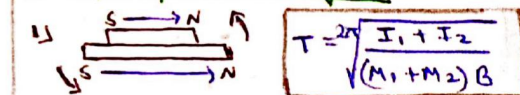
SHM of bar Magnet in Uniform MF :



$$T = \sqrt{\frac{MB}{I}} \Rightarrow T = 2\pi \sqrt{\frac{I}{MB}}$$

I → M.O.I of bar magnet
M → Magnetic moment

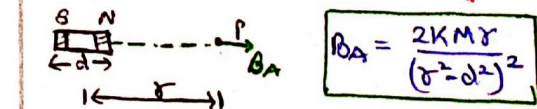
SHM of Coupled bar magnets :



→ M.O.I will always be added.

→ Magnetic moment net, will depend on orientation.

Magnetic Field due to Bar Magnet :



$$B_E = \frac{K M}{(r^2 + d^2)^{3/2}}$$

Magnetic Field due to Magnetic Dipole :

→ A bar magnet of very small length can be considered as a magnetic dipole. ($r \gg d$)

$$B_A = \frac{2KM}{r^3}$$

$$B_E = \frac{KM}{r^3}$$

$$B_r = \frac{2KM \cos \theta}{r^3}$$

$$B_\theta = \frac{KM \sin \theta}{r^3}$$

$$B_p = \sqrt{B_r^2 + B_\theta^2} = \frac{KM}{r^3} \sqrt{1 + 3 \cos^2 \theta}$$

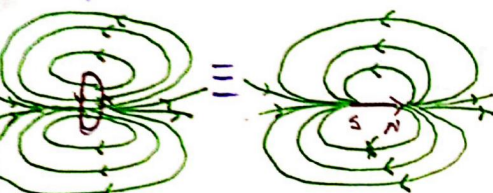
Also angle of B_p can be given by

$$\alpha' = \tan^{-1} \left(\frac{B_\theta}{B_r} \right)$$

$$\alpha' = \tan^{-1} \left(\frac{1}{2} \tan \theta \right)$$

Small Coil as Magnetic Dipole :

→ Very small current carrying coil produces MF similar to a magnetic dipole.



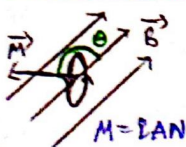
$$\vec{M} = n \vec{A}$$

$$\vec{M} = m \vec{d}$$

From this analogy, we can write $\vec{B}$ due to small coil anywhere in 3D space, considering it a dipole

$$\begin{cases} B_r = \frac{2K(2\pi R^2) \cos \theta}{r^3} \\ B_\theta = \frac{K(2\pi R^2) \sin \theta}{r^3} \end{cases}$$

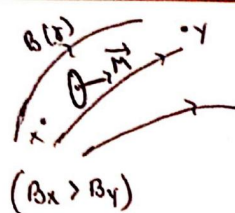
Force and torque on a coil :



$$F_{net} = 0$$

$$\tau_{net} = MB \sin \theta$$

→ If the magnetic field is variable, the net force on coil is given by



$$\vec{F} = m \frac{dB}{dr} \hat{r}$$

(along $\hat{r}$ dirⁿ)

Force between two magnetic dipoles :

$$M_1 \rightarrow F_B \leftarrow M_2$$

$$B_1 = \frac{2KM_1}{r^3}$$

Force on M_2

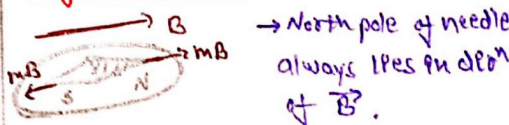
$$F = M_2 \frac{dB_1}{dr} = M_2 \left(\frac{-6KM_1}{r^4} \right)$$

$$\Rightarrow F_{dipoles} = \frac{6KM_1 M_2}{r^4}$$

Attractive Force

$$F_B \propto 1/r^4$$

Magnetic Needle :



→ Behave as a Magnetic dipole / light small magnet.

And defⁿ of magnetic needle at origin.

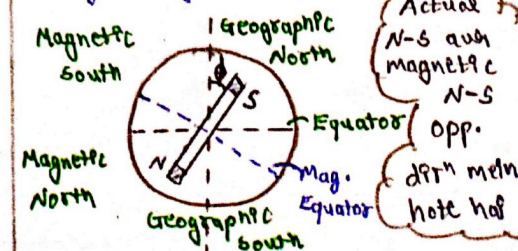


$$\begin{cases} B_1 = \frac{2KM_1}{r^3} \\ B_2 = \frac{2KM_2}{r^3} \end{cases}$$

$$\text{So, } \tan \alpha = \left(\frac{B_2}{B_1} \right) \Rightarrow \alpha = \tan^{-1} \left(\frac{M_2 B_1}{M_1 B_2} \right)$$

Earth Magnetism :

→ Due to rotation of charges in magma in core of earth, earth has its own magnetic field



Magnetic Meridian : Plane containing magnetic north and south poles

Geographical Meridian : Plane containing geographical north & south poles.

Angle of declination (ϕ) : Angle b/w both meridians. (ϕ lies b/w $12^\circ - 2^\circ$)

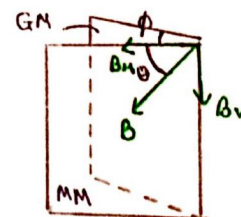


MM → Magnetic Meridian
GM → Geographical Meridian

Angle of dip (θ) : (Northern hemisphere)

(i) Angle made by resultant $\vec{B}$ of earth wrt horizontal of MM.

(ii) Angle made by freely suspended mag. needle wrt horizontal of MM



NOTE : In Southern hemisphere, θ is called angle of inclination, though its definition/meaning remains same

$$\rightarrow \tan \theta = \frac{B_v}{B_H} \quad \& \quad B = \sqrt{B_v^2 + B_H^2}$$

→ At Magnetic poles ($\theta = 90^\circ$)

$$B_H = 0 \text{ (min)} \quad B_v = B \text{ (max)}$$

→ At Magnetic equator ($\theta = 0^\circ$)

$$B_H = B \text{ (max)} \quad B_v = 0 \text{ (min)}$$

→ For earth's magnetic field

ϕ → Decides plane in which $\vec{B}$ lies at any place

$\phi \& \theta$ → Decides dirⁿ of $\vec{B}$

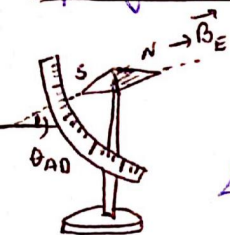
$\theta \& B_H$ → Decides magnitude of $\vec{B}$

Apparent Dip angle of a point on Earth surface:

→ Dip needle is an apparatus used to measure dip angle (θ) of earth's MF.

→ But, it measures correct θ , only when needle placed in MM.

→ If needle is placed at an angle α to MM, then it gives us apparent dip angle (θ_{AD}).



So here,

$$B_H = B_H \cos \alpha$$

$$\tan(\theta_{AD}) = \frac{B_V}{B_H \cos \alpha}$$

→ So relation b/w θ_{AD} and θ is

$$\theta = \tan^{-1} [\cos \alpha \cdot \tan(\theta_{AD})]$$

Oscillations of Compass Needle:



→ Normally, magnetic needle lies along B_H .

angular

→ If needle is given very small displacement θ , then

$$T = 2\pi \sqrt{\frac{I}{MB_H}} = 2\pi \sqrt{\frac{I}{MB \cos \theta}}$$

($I \rightarrow$ M.O.I of needle)

Since magnetic needle can be treated as a small magnet, we directly used expression for time period.

Magnetic Field Vectors $\vec{H}$, $\vec{I}$ & $\vec{B}$:

(I) Magnetising Field / Magnetic Intensity ($\vec{H}$):

→ It is the external field under which a material is placed for magnetisation

→ For example, if an iron core is placed inside a solenoid for magnetisation;

then mag. field responsible for its magnetisation is given by $\vec{H}$.

$$B = \mu_0 n i$$



Also,

$$H = n i \quad (n = \frac{N}{L})$$

$$B = \mu_0 H$$

(II) Intensity of Magnetization ($\vec{I}$):

→ When a magnetic material is placed in magnetising field then induced dipole moment per unit volume is $\vec{I}$.

→ For example, the dipole moment induced in above iron core after its magnetisation.

$$\vec{I} = \frac{\vec{M}}{V} \quad (V \rightarrow \text{Volume of material})$$

(III) Magnetic Susceptibility (χ_m):

→ It represents the ease with which a magnetic material can be magnetised.

$$\chi_m = \frac{I}{H} \quad (\text{It is scalar with no unit and dimension})$$

(IV) Magnetic Permeability (μ_r):

→ It measures the degree to which a magnetic material can be permeated by MF lines.

$$\mu_r = \frac{B_m}{H}$$

($B_m \rightarrow$ Total MF inside material)

→ Relation b/w μ_r & χ_m

$$\mu_r = (1 + \chi_m)$$

(For vacuum, $\chi_m = 0$)

Classification of Magnetic Materials:

→ On basis of magnetic properties, Faraday divided these materials into 3 categories.

(I) Diamagnetic:

$$\rightarrow \mu_r < 1 \text{ and } \chi_m < 0 \text{ (-ve)}$$

→ Magnetism due to orbital motion of electrons.

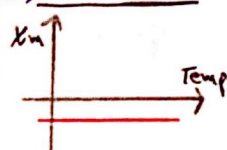
→ In ext. magnetic field, poor magnetisation in opp. dirⁿ.



$$B_m < B_0$$

→ Atom do not have permanent magnetic moment.

→ χ_m -T curve $\Rightarrow$



$\chi_m \rightarrow$ small, -ve, independent of Temp.

$$\chi_m \propto T^0$$

→ I-H curve $\Rightarrow$



$$\therefore \chi_m = I/H$$

$$\Rightarrow I = \chi_m \cdot H$$

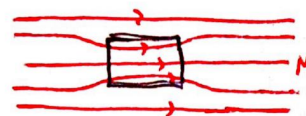
$\chi_m < 0 \Rightarrow$ -ve slope

(II) Paramagnetic:

$$\rightarrow \mu_r > 1 \text{ and } \chi_m > 0 \text{ (small +ve)}$$

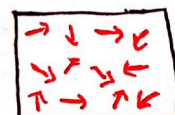
→ Magnetism due to spin motion of electron

→ In ext. magnetic field, poor magnetisation in same dirⁿ.



$$B_m > B_0$$

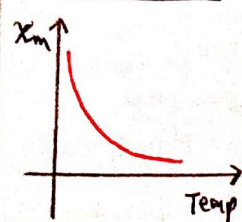
→ Atoms have permanent mag. moment which are randomly oriented.



In absence of B_{ext},

$$M_{net} = 0$$

→ χ_m -T curve $\Rightarrow$



$\chi_m \rightarrow$ small, +ve, varies in-versely with temp

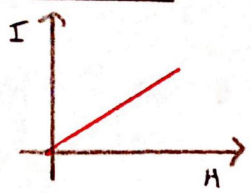
$$\chi_m \propto \frac{1}{T}$$

[Curie Law]

If $T \uparrow \Rightarrow \chi_m \downarrow$
If $T \downarrow \Rightarrow \chi_m \uparrow$

$$\chi_m = \frac{C}{T} \quad \text{Curie Const.}$$

→ I-H curve ⇒



∴ $\chi_m = \frac{I}{H}$
 $I = \chi_m H$
 → small +ve slope

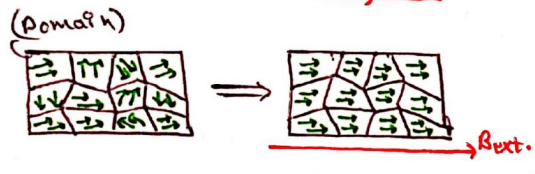
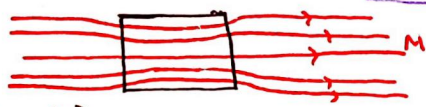
(III) Ferromagnetic ⇒

→ $\mu_r \gg 1$ and $\chi_m \gg 0$ (big +ve)

→ Magnetism due to formation of domains.

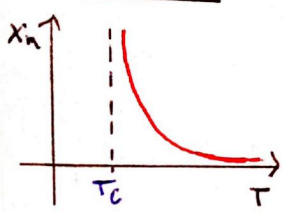
Domain is a region inside material where groups of mag. moments naturally align in same dirⁿ.

→ In ext. MF, strong magnetisation in same dirⁿ. $B_m \gg B_0$



→ Atoms have permanent magnetic moment which are organised in domains.

→ χ_m -T curve ⇒

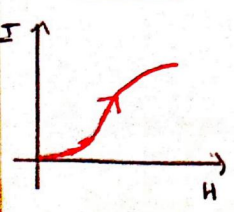


χ_m → Very large, positive, temp dependent.

$\chi_m \propto \frac{1}{T - T_c}$ (For $T > T_c$)
 T_c → Curie Temp.
[Curie Weiss law]

(Temp. after which χ_m starts decreasing & ferromagnetic material starts converting into paramagnetic material)

→ I-H curve ⇒

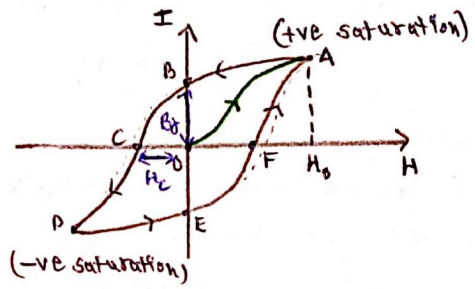


→ I is very large, +ve & varies non linearly with field.

$\chi_m = \frac{I}{H}$ Not valid for Ferromagnetic

Magnetic Hysteresis :

→ A lag of magnetic flux density (B) behind the magnetic field strength (H) is 'Magnetic Hysteresis'.



D → A : On TH, I↑ and gets saturated for $H = H_0$.

But the T in I is not immediate and lags behind the T in H.

A → B : On ↓H to zero, still I is retained due to lag.

OB is called 'RETENTIVITY'
 (y-intercept of I-H curve)

B → C : On TH more in -ve dirⁿ, I becomes zero not at $H=0$ but $H = \text{some -ve}$ due to lag.

The value of 'H' at which $I=0$ is called 'COERCITIVITY' (OC)

C → D : Further TH in -ve, I gets saturated at D, and the loop continues.

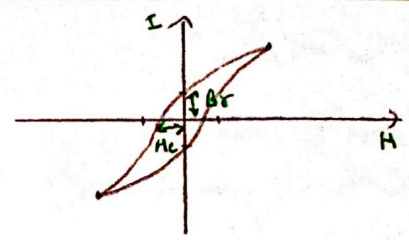
NOTE: Area of loop ABCDEFA shows thermal energy / heat produced per vol. in 1 cycle

→ Ferromagnetic materials can be categorised into two types :

1) Soft Magnetic Materials ⇒

→ Low Retentivity, low coercivity & low hysteresis loss.

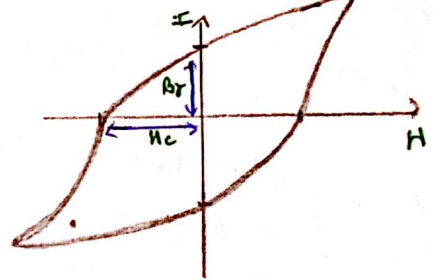
→ Suitable for electromagnets & cores of transformers.



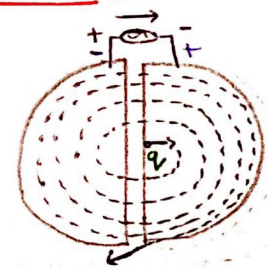
2) Hard Magnetic Materials ⇒

→ High retentivity, high coercivity and large hysteresis loss.

→ Suitable for permanent magnets. (Example: steel, Alnico)



Cyclotron :



⊙ B

As $v \uparrow$
 $\Rightarrow r \uparrow$

→ Cyclotron is a device used to produce accelerated charges.

→ It consists of 2 metallic D's, connected by AC source which change their polarity after each revolution of charge

(i) Frequency of AC source

$f = \frac{1}{T} = \frac{qB}{2\pi m}$

(ii)

$R_{max} = \frac{mv_{max}}{qB}$

(iii) Charge accelerates two times per one complete revolution. So

In 1 rev, $a = 2 \times qV$ ($V \rightarrow$ Potential diff)